

ColorNetVis: An Interactive Color Network Analysis System for Exploring the Color Composition of Traditional Chinese Painting

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Abstract—In the field of digital humanities, color research aims to discover explanations for painting history and color usage habits. However, researchers analyzing color relationships is challenging and time-consuming, as it requires color extraction and a detailed review of many painting images for reference and comparison of color relationships. In our work, we propose ColorNetVis, an interactive color network analysis tool that enables researchers to explore color relationships through color networks. The core of ColorNetVis is a bipartite network model that establishes a bipartite relationship between colors and Chinese painting within a scope based on color difference measurement. It constructs a one-mode color network through projection algorithms and similarity calculation methods to discover the relationship between colors. We propose a coordinated set of views to demonstrate the combination of determined color networks with painting types and real-world attributes. We use color space view, color attribute distribution view, and single color query components to assist researchers in conducting detailed color analysis and validation. Through case studies, researcher reviews, and user studies, we demonstrate that ColorNetVis can effectively help researchers discover knowledge of color relationships and potential color research directions.

Index Terms—Color analysis, color network, traditional chinese painting, visual analytic.

I. INTRODUCTION

THE color culture [2], [30] of traditional Chinese painting is one of the most precious treasures of contemporary times, and its color composition is a profound and complex issue

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that involves traditional Chinese aesthetics [9], philosophy [13], culture [12], and artistic concepts [32]. The color research of Chinese painting has rich value in digital humanities, providing critical visual dimensions and clues for a deeper understanding of culture, history, and society [12], [14], [49].

Color analysis [6], [7] is an important research field in current digital humanities research, which can inspire explanations of painting history and color habits. Digital humanities provide a platform for interdisciplinary cooperation and exchange [20], [21], [25], allowing researchers from different fields to conduct in-depth discussions and research on the theme of color, thereby promoting the development of related fields [36], [46], [49]. Through color analysis and digital technology, it is possible to provide a more accessible display and experience of Chinese painting, thereby promoting the recognition and understanding of Chinese painting among different populations. This new concept has sparked widespread discussion in the digital humanities community in China, examining the colors between Chinese paintings from a data perspective [23], [29]. More importantly, it inspired subsequent works to use color analysis methods to explain known colors and propose new concepts. The current color analysis research mainly focuses on artistic or aesthetic interpretation and numerical analysis, focusing on a color group and its relationship. For example, the color scheme of mountains is mainly cyan. At the same time, trees are primarily black, so there is a close connection between cyan and black in paintings of mountains and trees.

The color network visual analysis [16], [22], [24], [33], [38], [49] method can discover the relationships between colors from large-scale image data, but more is needed. Existing color network visual analysis focuses on color extraction and analysis based on clustering, which performs well in color-dense areas. At the same time, the results of these clustering methods are unreliable for Chinese paintings with diverse embellishments. Besides, the construction method of color networks is based on the frequency of color occurrences, which can lead to colors with fewer occurrences being ignored or filtered. Moreover, researchers must repeatedly compare and reference image files to discover and verify color relationships, which is very time-consuming. Therefore, a comprehensive model that combines color and painting is needed, and the exploration process of integrating domain knowledge and expert research goals into an intuitive visual interface is highly desirable. To this end, we

worked closely with digital humanities researchers engaged in color research in Chinese painting, observing their behavior and needs, and identified two challenges:

- Researchers must invest significant time in synthesizing and cross-examining the relationships between various colors in a large amount of image data. This process is very time-consuming because Chinese painting involves a wide range of content, covering various attributes such as materials, shapes, and rendering methods.
- The process of discovering color relationships is repetitive because they must constantly query and search multi-attribute and similar color combinations. They usually start with a familiar color, discover interesting color forms, and recursively explore and study related colors.

Considering their exploration process and these obstacles, it is necessary to effectively analyze the color relationship of Chinese painting and provide visual-driven solutions. The core of our solution, ColorNetVis, is a bipartite network model that can establish the relationship between color and painting. Specifically, given an initial analysis range of interest, it constructed a color-painting bipartite network using color data from paintings in the database. It constructs color nodes using color difference measurement. Based on the projection algorithm of the bipartite network, it converts the color-painting bipartite network into a one-mode color network, discovering the hidden relationship between colors through the centrality index of the network. In the integrated visual interface, intuitively investigate the color relationships of paintings from multiple perspectives (such as shape and material). Researchers can explore real-world attributes and similar colors or paintings of interest through single color queries. We have demonstrated through actual case studies and expert interviews that ColorNetVis frees researchers from the heavy workload and improves analytical performance, verifying the effectiveness of the proposed visual analysis method. On the whole, this work makes the following contributions:

- We have summarized the task flow and design requirements for exploring multi-dimensional color patterns of Chinese painting.
- We propose a color network analysis method that utilizes a color-painting bipartite network to quantify and explain the complexity of color composition in Chinese painting from a macro perspective.
- We have developed an interactive system, ColorNetVis, which can free researchers from extracting, querying, and validation color pattern.

II. RELATED WORK

This section reviewed the relevant work on color-driven digital humanities analysis and color network visual analysis research.

A. Color-Driven Digital Humanities Analysis

The color-driven digital humanities aims to optimize the analysis process through color coding to achieve visual analysis and digital humanities research [28]. For example, Thudt et al. [35] designed a digital library system called The Bohemian Bookshelf, which uses color coding to facilitate the analysis

of exciting books that users accidentally discover while using digital libraries. Sari et al. [31] proposed an interactive automatic visualization system based on a database of many painting works. The system locates portrait works and automatically categorizes them by gender, then segments clothing areas and finds their dominant colors, allowing users to explore trends in clothing color changes. Kaneko et al. [18] proposed a multidimensional data visualization technique that visualizes the image features and classification information of painting works and highlights association rules so that users can explore the painting works of interest. Yang et al. [40] used clustering and color space visualization methods to quantitatively analyze colors and explore the color combination patterns of Peking Opera facial makeup. Wu et al. [37] proposed an adaptive method for transferring colors from arbitrary reference images to vector maps to facilitate the production of attractive and practical maps and visualizations. Stepanova et al. [33] established a connection between human aesthetic evaluation of artworks and their visual characteristics. By extracting and analyzing the color composition of paintings sold at auctions, they found that paintings with a high diversity of colors carry a more premium than monochromatic style. Zhu et al. [49] designed an intelligent analysis system for image color to achieve automatic classification of different color regions in Chinese painting, ultimately allowing for the judgment of the most likely pigments to be used for each color. Flueckiger et al. [7] focused on a theoretical and analytical concept of film color, linking style, expression, and narrative dimensions with developing and evaluating digital methods, aiming to provide a more comprehensive approach to analyzing various aspects of color in movies. ChortVA is a queue analysis tool proposed by Zhang et al. [46] for historical figures, which enhances features by using shape, color, and thickness to enhance differentiation and user perception. In short, color-driven digital humanities analysis is of great significance. It can be used for in-depth analysis and research on text [35], images [18], [31], [37], [49], historical and cultural heritage [33], [40], [49], social phenomena [7], [37], and other aspects. Color analysis in Chinese painting plays a positive role in promoting world painting development and improving artists' creative level. We propose a color network analysis model based on previous research and conduct color analysis on Chinese painting. We aim to provide a reliable analysis model and expand color-driven digital humanities research methods.

B. Visual Analysis of Color Networks

The color network visual analysis method has a wide range of application scenarios in the field of digital humanities. Liu et al. [24] extracted a series of feature colors from the Chinese pattern library through two clusters, established a color-matching network between feature colors, and recommended the color scheme that best reflects the cultural essence for designers. Li et al. [22] developed a color-matching design system based on a color adjacency network model, which assisted in the critical steps of color-matching design from color extraction to reuse for automotive exterior color matching. Xu et al. [38] developed a color network model for generating

and optimizing color schemes and verified the effectiveness and feasibility of the model by transferring color schemes to traditional pattern images. Cao et al. [16] analyzed the color and color matching mechanism of ethnic clothing by establishing a color relationship network model to analyze the form of color composition and the internal correlation of each ethnic group. Yu et al. [41] constructed a machine learning based color network model for the folk painting, which can accurately reflect the color characteristics and color-matching design logic of Ansai folk painting. Feng et al. [5] used color network models to represent the relationships between pixel color information to obtain color themes with a wide range, appropriate quantity, and high accuracy. Xu et al. [39] constructed a primary color network relationship model to visually represent the color mechanism of clothing and assist in clothing planning and design decisions. The above research indicates that the scope of color network visualization analysis involves art analysis, cultural heritage protection, historical research, and visual anthropology. However, most of the above studies are based on clustering methods. For Chinese painting, the proportion of embellishment colors is relatively small, and the main colors obtained through clustering methods cannot be accurately analyzed for the color composition form of Chinese painting. Therefore, we propose an improved bipartite network method to construct a color network to ensure complete visualization of low-proportion colors.

III. BACKGROUND

In this section, we introduce the background information of color research in Chinese painting and outline the requirements obtained. We have worked closely with five experienced researchers over the past two years (E1-E5). Three (E1-E3) are professors engaged in color research in Chinese painting. One is mainly engaged in historical literature research, and the other two are mainly engaged in color aesthetics research in Chinese painting. The other two (E4 and E5) are doctoral students, mainly engaged in the exhibition of painting works and research on traditional Chinese art. They all have practical experience in color research in Chinese painting and unanimously believe it is an essential component of Chinese painting research. We observe and document their process of color analysis for a long time, hold regular meetings or interviews with them, ask them to describe different problems in color research, discuss system design, and evaluate the final system. Based on the meeting and interview records with the expert group, we have determined their general process for studying the color of Chinese painting, as shown in Fig. 2. Our collaboration includes five stages: data collection (Section III-A), task analysis (Section III-B), method description (Section IV), system design (Section V), and system evaluation (Section VI).

A. Data Description

In this work, we use the traditional Chinese painting color (TCPC) database to study painting colors in Chinese history. The database is based on *A Comprehensive Collection of Ancient Chinese Paintings* [47] and is a large-scale, historical, and cross-ethnic major cultural project. TCPC database has been

TABLE I
DATA FORMAT OF TCPC DATABASE

Data format	Explanation
ID	Index of paintings in the database.
Name	The Chinese name of the painting.
Time	Validated painting creation time, using a time period such as the Tang and Song dynasties.
Theme	The theme of a painting depends on the content, mainly including landscape painting, flower-and-bird painting, figure painting, and other themes.
Material	Material refers to the base used in the creative process of a painting, including paper, silk, and linen.
Rendering method	The use of colors in the painting, including heavy coloring, light coloring, and line drawing.
Shape	The shapes include book leaves, page with a pivot, hand scrolls, and sectors, representing the storage form of Chinese painting.
Color values and labels	Hexadecimal data and proportion of painting colors, stored in the form of color values and color labels.

highly valued and continuously followed for many years and contains 12405 Chinese paintings made of materials such as paper and silk from 263 cultural institutions around the world from the pre-Qin to the Qing Dynasty, covering most of the painting treasures. TCPC database has been recorded and confirmed by experienced researchers. We first preprocessed 560 representative paintings from the Tang and Song dynasties in the database, removing irrelevant borders and text parts while retaining the main parts, especially the colorful areas. We use Python's extcolors package (<https://pypi.org/project/extcolors/>) to extract colors from painting data within the 20 tolerance and format data obtained, as shown in Table I. The extracted colors with a proportion between 0.05 and 80 are stored as color values and labels.

B. Requirement and Task Analysis

Presenting the colors of a large amount of data to researchers in an easily understandable form is challenging. We interviewed experts in the study of Chinese painting colors to understand their traditional workflow and guide our system design. The traditional color research is shown in Fig. 2(a). Experts point out that the traditional analysis process is energy consuming and time-consuming. We summarized two main system requirements (SR1 and SR2). Subsequently, we decomposed SR1 and SR2 into six design tasks (T1-T6), enabling researchers to have broader color analysis and deeper color investigation capabilities to adapt to their challenges.

SR1. Provide visual explanation for the color relationship in Chinese painting: The system should summarize and visualize color relationships from a large-scale image database. To find the target range, researchers need to browse and filter the painting

image data. Then, they must conduct heuristic searches based on their experience and goals. To verify the appearance of color with other colors, researchers need to compare the relevant paintings repeatedly. Therefore, the system should provide practical visual explanations of color relationships based on users' interests and goals.

- *T1 Specify initial research scope:* Researchers' interests come from different perspectives. Chinese painting is a vibrant, diverse, and historical art form involving many themes, techniques, and historical backgrounds. Determining research scope can make research more focused, targeted, and efficient. The system should specify the initial range in multiple ways to support research directions for different painting types.
- *T2 View the distribution of colors:* Studying color distribution also helps researchers compare the similarities and differences between themed Chinese painting works. For example, paintings with different themes and materials from the same historical period may have different color distribution, reflecting changes in culture, aesthetics, and style.
- *T3 Analyze a predefined set of color relationships:* In Fig. 2(a), researchers typically need to determine a set of colors with potential relationships to be analyzed, making their research more focused and targeted. Predefining a set of colors can also prevent the scope of colors from being too broad, making the research more in-depth and detailed. Simultaneously, Researchers can analyze these colors more accurately, which helps to understand and explain the use and distribution of colors in Chinese painting, reducing the possibility of misreading or misunderstanding.

SR2. Support interactive and recursive exploration of color patterns: In addition to providing an overview of the relationships between colors, the system should allow researchers to interactively and recursively explore the relationships between multi-attribute colors and utilize the color network to discover implicit color patterns.

- *T4 Single color query:* Researchers need to understand the use, significance, and expressive effects of a single color in Chinese painting through detailed research in a wide range of color spaces. Single color query function helps researchers have a more comprehensive understanding of the characteristics and connotations of Chinese painting. By comparing the use of this color in different paintings, researchers can better understand the meaning and role of color in different contexts.
- *T5 Analysis of the overall importance of colors:* When representing the relationship between colors in network visualization, the network attributes of color nodes have practical significance, so necessary explanations, and analysis should be provided to help traditional researchers better understand the deep-level relationships in the network. The system should highlight the importance of different colors based on the network centrality indicators of each color node.
- *T6 Timely update of color network:* The system should have real-time updates of color network and node centrality

indicators based on the research scope to adapt to the researcher's research scope changes.

IV. METHOD DESCRIPTION

This section provides an overview of visual analysis methods for visualizing Chinese painting colors through color networks and introduces the network models used. The network model is divided into two parts. The first part establishes a bipartite network from the database based on the specified paintings and ranges. The second part establishes a one-mode color network through the bipartite network, as shown in Fig. 3. All networks are generated based on formatted data in Table I.

A. Establishment of Color-Painting Bipartite Network

A bipartite network [10], [15], [48], [50] describes the relationship between two objectives, such as papers and authors. A bipartite network can be represented as $G = (U, V, E)$, where U and V represent different node sets, and E represents edges between nodes belonging to U and V . This work represents the bipartite network as the relationship between the painting and the corresponding colors. When the researcher determines the research scope (Fig. 3(b1)), the system will screen all paintings and their color data that meet the researchers' scope from the color library. Each painting can establish a single painting bipartite network based on the attributes in the database. For painting i , its single painting bipartite network can be represented as $G_i = (U_i, V_i, E_i)$, where $U_i = (u^i)$ represents the painting itself, and $V_i = (v_1^i, v_2^i, \dots, v_k^i)$ represents the color set of the painting i (Fig. 3(a2)). We merge all single painting bipartite networks through color difference measurement ΔE to obtain the color picture binary network within the entire research scope, represented as $G = (U, V, E)$. In the color-painting bipartite network, $U = (U_1, U_2, \dots, U_m)$ represents the set of m paintings, $V = (V_1, V_2, \dots, V_n)$ represents the merged set of n color nodes, and E represents the edges of U and V (Fig. 3(a3)). When constructing a color-painting bipartite network within the research scope, color difference measurement ΔE is the key to merging similar colors. In literature, $3 \leq \Delta E \leq 7$ is usually used as the human eye's distinguishable range [34], [42], and increasing the color difference ΔE will merge more similar color nodes into the same color node, reducing the number of color nodes. Therefore, we used the CIELAB2000 color difference measurement [27] as a calculation criterion. Researchers can provide an overview of the relationship between color and painting through the overall color painting bipartite network.

B. Establishment of One-Mode Color Network

Projecting a bipartite network onto one side is a standard analysis method [10], [48]. A one-mode color network can be obtained by projecting the color painting bipartite network towards the color side, where each node is a color. One-mode color networks can provide new insights for researchers to analyze color relationships through node centrality analysis methods. However, after the bipartite network projection, weights need to be calculated. We propose using cosine similarity to calculate

the edge weights of two color nodes, as shown below.

$$Sim(V_i, V_j) = \frac{\sum_{k=1}^m x_{ik}x_{jk}}{\sqrt{\sum_{k=1}^m x_{ik}^2} \sqrt{\sum_{k=1}^m x_{jk}^2}} \quad (1)$$

where m represents the number of paintings within the research scope. If color V_i does not exist in painting k , $x_{ik} = 0$, else $x_{ik} = 1$. Similarly, if color V_j does not exist in painting k , $x_{jk} = 0$, else $x_{jk} = 1$.

Cosine similarity measures the degree of similarity between two colors as samples of a painting, indicating a greater likelihood of these two colors appearing in the same painting (Fig. 3(a4)). One-mode color networks can analyze more properties through traditional network analysis methods, so we analyze the importance of color in the network through degree centrality and eigenvector centrality (Fig. 3(b2)).

V. SYSTEM DESIGN

We developed an interactive prototype system to assist researchers in implementing the methods mentioned in Section IV. This section provides an overview of the system and details of visual design and interaction.

A. Color Analysis Task Flow

We have developed a visual analysis interactive prototype system, ColorNetVis. ColorNetVis follows a two-stage color relationship analysis workflow, as shown in Fig. 2(b). To establish the research scope (T1), the *scope specification component* (Fig. 3(b1)) supports researchers to specify an initial range based on their research interests, including theme, material, rendering method, and shape, and then import the initial range into the workflow to retrieve the corresponding painting and color attributes in the TCPC database. Given a defined range, ColorNetVis uses a color library to establish a color painting bipartite network within the range, forming a color relationship overview (T2). Researchers use the color painting bipartite network and one-mode color network in *network visualization view* to discover color composition and color relationships in the painting. Through the color one-mode network, the overall importance of colors in the network can be viewed through the *color node attribute view* (T5), as shown in Fig. 3(b2). Researchers can filter a group of colors of interest, find the similarity and attribute distribution (T3) between these colors and other associated colors through the node heat map and *color attribute distribution view*, and adjust the color edge weight in the single-mode color network to further reveal the strong relationship of colors (T6), as shown in Fig. 3(b3). After the overview analysis, researchers can select and focus on one or more colors for further analysis. Researchers can use *color search component* to discover more information about colors and shorten the time spent cycling through painting images and colors. Researchers can freely query any color, which can be in the color network or in the *color space view*, and generate similar color nodes in the color network and attribute (T4) in the corresponding painting, as shown in Fig. 3(b3). In addition,

researchers can discover the single painting bipartite network of interest through the painting query function.

B. Scope Specification Component

The *scope determination component* (Fig. 1(a)) provides a control panel for data queries to help researchers quickly locate the color range they want to study and generate network visualization. Supports five separate and combined query methods: 1) Query based on the name of the painting (the query result is a single painting bipartite network); 2) Theme-based query; 3) Rendering method based query; 4) Material-based query; 5) Shape-based query.

C. Network Visualization View

The *Network visualization view* (Fig. 1(b)) consists of two parts: 1) Color painting bipartite network and monochromatic color network visualization view (Fig. 1(b)), and 2) Color heat map analysis view (Figs. 1(b-1) and 4(b)). These two parts jointly support the analysis of color relationships in Chinese painting. *Network visualization view* using the network construction method described in Section IV and encoding each node based on the color value of the color itself. Researchers can click on the color node to view the RGB value and node centrality. The network layout adopts the FR layout [8] to display the clustering effect as much as possible. Researchers can view a one-mode color network through the transform button and reduce the entire network by adjusting weights to explore colors with different relationship intensities. When researchers need more distinguishable and analyzable colors, they can adjust ΔE to reduce the number of nodes in the network. Researchers can click and add a set of colors of interest one by one and discover the relationships between colors in a more focused and clear form through node heat maps. In the node heat map analysis view, the heat map is generated through a one-mode color network, which displays the top 15 other colors strongly related to the colors defined by the researcher, as shown in Fig. 4(b).

Justification. In our proposed method for constructing color networks, increasing ΔE will result in fewer color nodes. Taking the one-mode color network of the Song Dynasty (with edge weights greater than 0.3) in the dataset as an example, we demonstrate the color network differences caused by ΔE , as shown in Fig. 5. Increasing ΔE will significantly reduce the number of color nodes. Although similar colors may decrease, they increase color distinguishability. On the contrary, a smaller ΔE leads to more color nodes to allow researchers to explore color relationships with more minor differences.

D. Color Node Attribute View

The *node attribute view* (Fig. 1(c)) consists of two parts: 1) Current node attributes and 2) Node centrality indicator statistics (Fig. 1(c)). The current node attribute view displays detailed information about the color, including name, RGB value, degree centrality, betweenness centrality (BC), closeness centrality (CC), and eigenvector centrality (EC). The statistics of network centrality indicators show the top 15 normalized rankings of

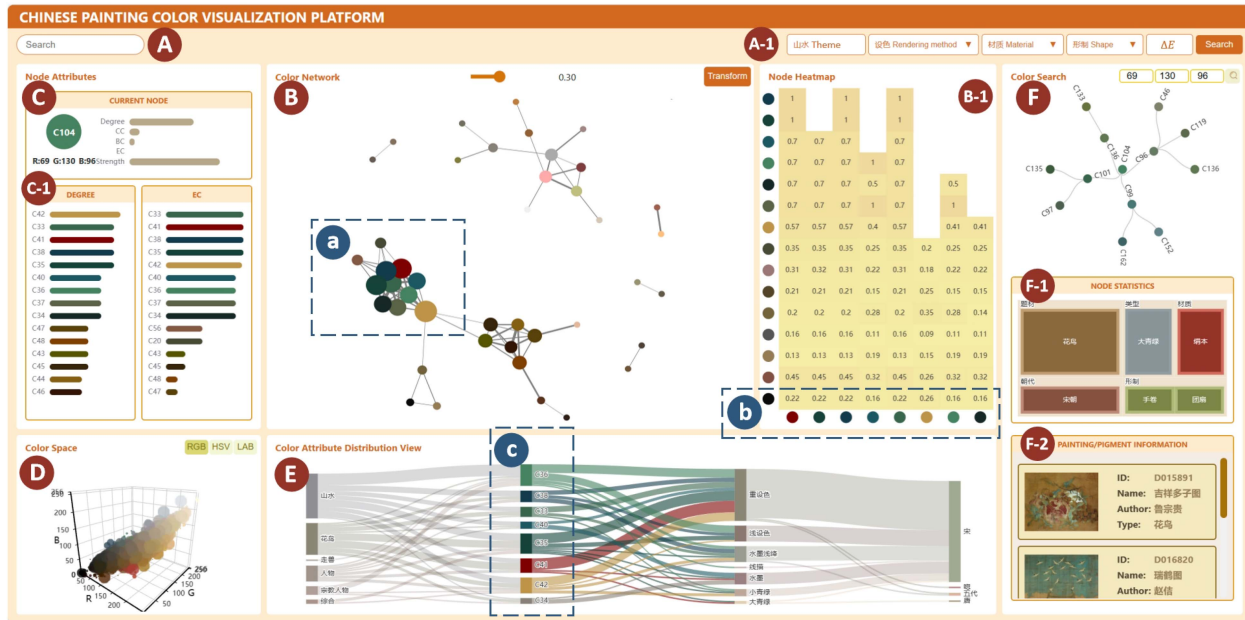


Fig. 1. ColorNetVis simplifies the workflow of color analysis in Chinese painting and provides visual explanations for the color-painting and color pair relationships. (A) Scope specification component, (B) Network visualization view, (C) Color node attributes view, (D) Color space view, (E) Color attributes distribution view, and (F) Color search component.

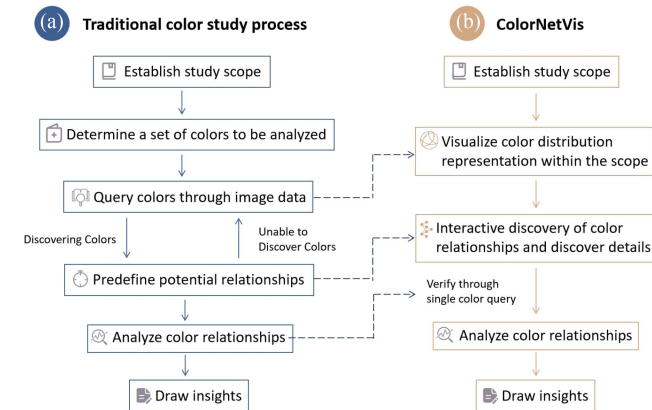


Fig. 2. A comparison between (a) traditional color research workflow and (b) proposed workflow applied in ColorNetVis.

degree centrality and EC of current network color nodes in the network (Fig. 1(c-1)). The color encoding of bar charts also uses color values of colors. We mainly use the following two node centrality to represent the network attributes of colors, with the aim of discovering colors with different meanings.

Degree centrality: Degree centrality [44] in network science represents the connections between a node and other nodes. Our color network model represents the number of combinations of color with other colors, reflecting the universality of colors.

Eigenvector centrality: Eigenvector centrality [1], [26] reflects the comprehensive importance of nodes in network science, aiming to conduct a comprehensive ranking of colors in color networks to reflect the importance of colors within the set research scope.

E. Color Space View

The *color space view* (Fig. 1(d)) shows the overall distribution of all colors in the TCPC database under the current research scope. Each extracted color is visualized in the color space and provides interactive functionality. Researchers can click on a color to view its RGB/HSV/LAB values and the corresponding painting and formatted attributes (Fig. 4(a)). Unlike the approximate calculation of colors in color networks, color space views provide researchers with the most primitive colors in the database and provide a foundation for color query functions.

F. Color Attribute Distribution View

The *color attribute distribution view* displays the path flow of colors in real-world attributes in a Sankey diagram (Figs. 1(e) and 4(d)), with each column summarizing the corresponding real-world attribute information of all colors from left to right. The first column from the left represents the theme of the painting, and the height of the block reflects the relative proportion of color in that attribute. The second column represents the colors in the database with color differences within the set value from predefined color nodes, the third column represents the rendering method, and the fourth column represents the dynasty to which these rendering methods belong.

G. Color Search Component

To help researchers search for colors of interest and discover their real-world properties, the *color search component* (Figs. 1(f) and 4(c)) supports researchers in searching and validating colors through a hierarchical model. Researchers can input any color RGB value and then match the color node in

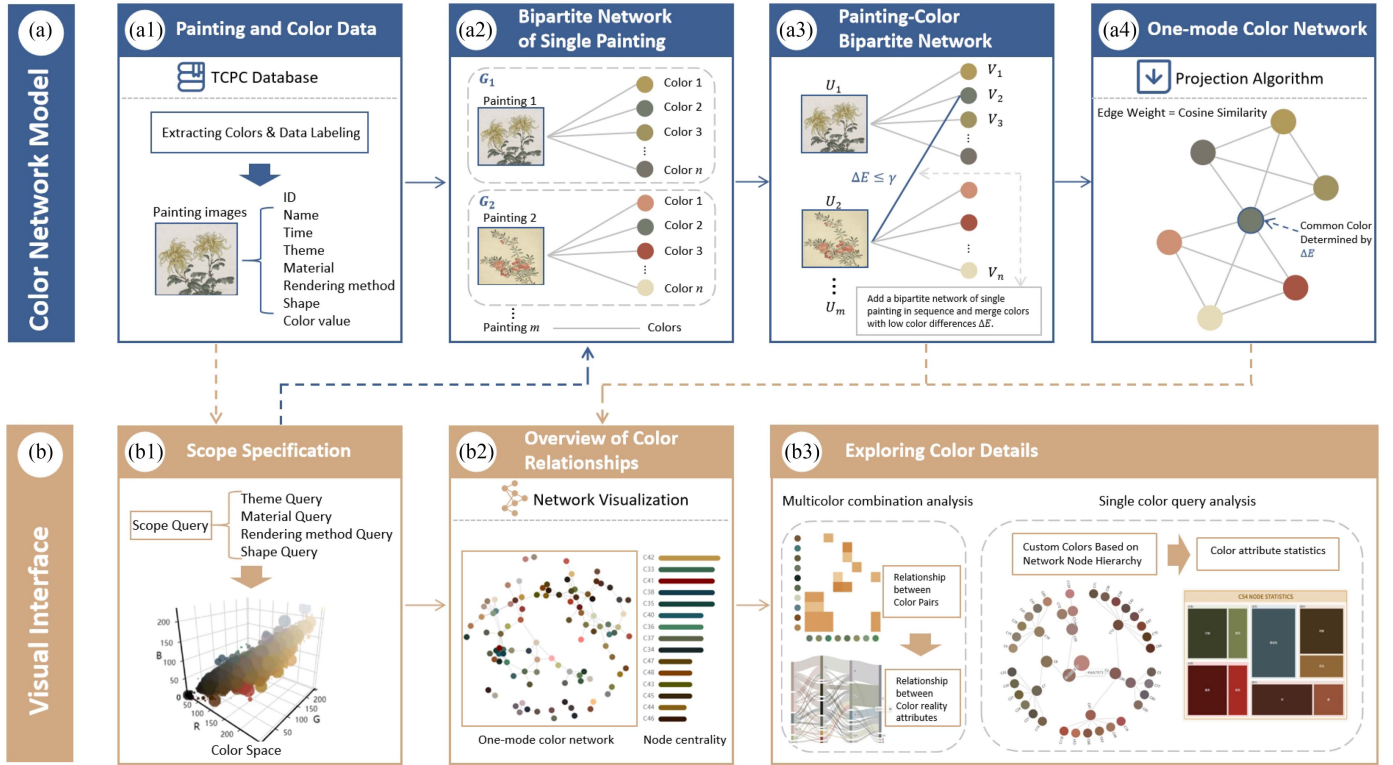


Fig. 3. ColorNetVis includes (a) a color network model and (b) a visualization interface. We first construct a painting data structure based on the TCPC database (a1). Here, examples of Painting 1 and 2 are used to help explain. Initially, a color-painting bipartite network (a2) is established based on the color values in the database. Similar color nodes (a3) are merged using color difference measurement ΔE [17]. Then, a one-mode color network (a4) is established using a projection algorithm. The generated color space and color network are presented in (b1) and (b2). At the same time, (b3) provides color combination analysis and single color query components for researchers to explain further and explore colors.

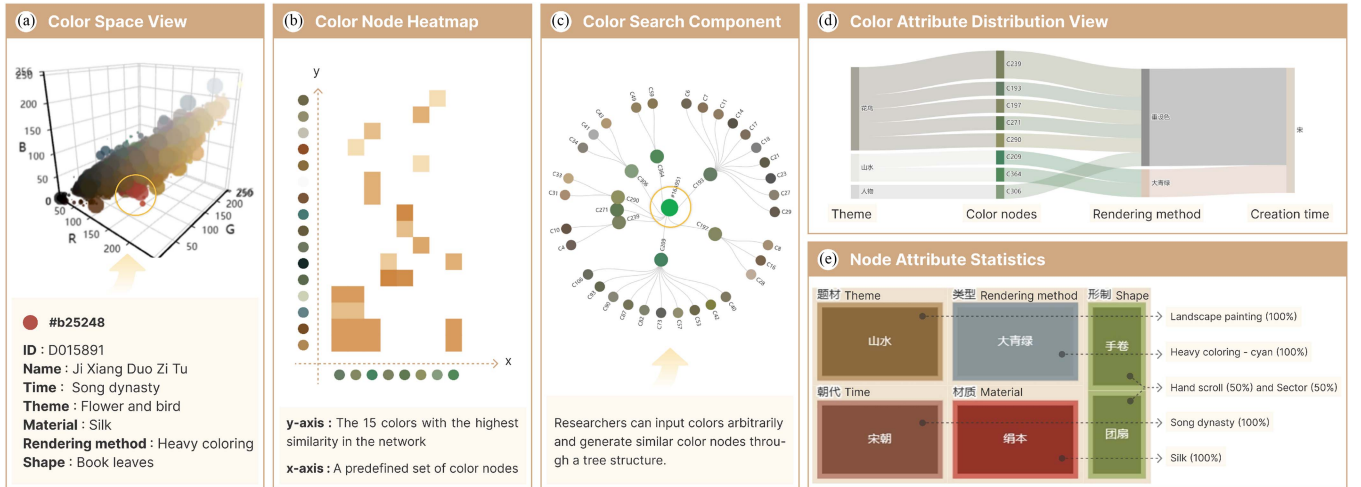


Fig. 4. Detailed description of different components of ColorNetVis. (a) Color space view, (b) Color node heatmap view, (c) Color search component, (d) Color attribute distribution view, and (e) Node attribute statistics.

the network with the smallest one as the root node. Afterward, color nodes with a color difference distance $\Delta E < 10$ from the previous layer of leaf nodes are searched in the color network, serving as the second and third layers of leaf nodes, ultimately forming a hierarchical structure composed of color nodes in the network. A hierarchical tree structure displays the similarity

relationships between colors in the database in an explicit manner. Through the node statistical attributes represented by the tree map (Figs. 1(f-1) and 4(e)), researchers can click on the color in the hierarchical structure to discover the actual attributes of that color. The painting display component is a supplement to the color query component, and researchers can display which

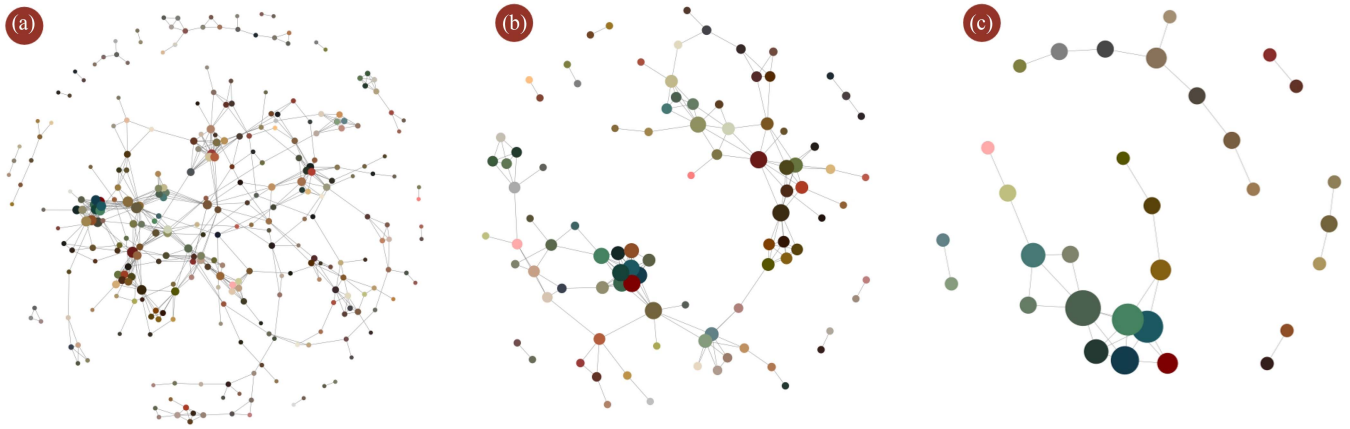


Fig. 5. One-mode color networks with different ΔE . (a) $\Delta E = 3$, (b) $\Delta E = 5$, and (c) $\Delta E = 7$.

paintings have that color by clicking on them and finding more painting information by dragging the slider (Fig. 1(f-2)).

VI. EVALUATION

We conducted two case studies and expert interviews to verify the effectiveness and usefulness of our ColorNetVis.

A. Case Studies

We invited the researchers mentioned in Section III to freely explore ColorNetVis based on their research intentions and interests. We encourage researchers to adopt a think-aloud protocol and record how they use our system, as described in the following cases.

Case1. Discovering Different Color Relationships from the Theme of the Painting: E1's research interest mainly lies in the theory of color composition in Chinese painting. He wants to examine the similarities and differences in the overall distribution of Song Dynasty landscape paintings, flower-and-bird paintings, and figure paintings. E1 conducted a study on the color composition of landscape, flower-and-bird, and figure paintings in the Song Dynasty using ColorNetVis.

With a clear goal, E1 initialized the database group with the *Song Dynasty* and *landscape painting* as constraints through the *scope determination component* (T1) and set $\Delta E = 7$. ColorNetVis returned a color-painting bipartite network with 288 nodes and 755 edges, as shown in Fig. 6(a). In this bipartite network, the central part tends to have dim colors, indicating that the overall color trend of landscape paintings in the Song Dynasty was towards simplicity and less color. The edge of the network reflects the color scheme style of Song Dynasty landscape painting. E1 noticed apparent green groups at the bottom of the entire network, which have higher saturation and are easier to distinguish than other colors in the bipartite network.

E1 discovered through the color painting bipartite network that these colors did not only appear in a single painting but in multiple paintings, which stimulated E1's research interest. Therefore, E1 converted the bipartite network into a one-mode color network using the transfer button (T6), filtered out edges below 0.3 using the weight axis, and obtained a one-mode color

network with 43 nodes and 86 edges, displaying the overall color distribution and relationship of Song Dynasty landscape painting (T2), as shown in the Figs. 6(d) and 1(a), for further analysis. In Fig. 6(d), blue-green has prominent clustering characteristics, and they are connected to other colors through node ● as intermediate node (Fig. 1(a)). In addition, through a one-mode color network, E1 also found that the color connectivity of landscape paintings in the Song Dynasty was weak. The entire color network was divided into two main connected components, one containing many heavy green colors and the other containing darker colors.

Through the *color node attribute view* (Fig. 1(c)), E1 found that nodes ●, ●, and ● were highly important in the color network of landscape painting (T5). To compare whether other painting themes in the Song Dynasty had these colors, E1 rebuild the color networks of flower-and-bird and figure painting through scope determination, as shown in Fig. 6(T6). In the bipartite network of Song Dynasty flower-and-bird paintings shown in Fig. 6(b), the number of color types is less than that of landscape paintings. The colors in the center of the network are relatively uniform. Overall, the saturation of colors is relatively low. One-mode color network visualization reveals that flower-and-bird painting (Fig. 6(e)) has a higher overall quality than landscape painting, indicating that the color combinations in flower-and-bird painting are more extensive. The essential colors in flower-and-bird painting are nodes ● and ●. In addition, in figure painting, the number of colors is less than that of landscape painting and flower-and-bird painting. The network's overall structure is looser than flower-and-bird painting, indicating fewer color combinations in figure painting. The essential colors in figure painting are nodes ● and ●, similar to flower-and-bird painting.

In the landscape painting color network, E2 wants to know which colors are associated with the color set (●, ●, ●, ●, ●, ●, ●, ●) in all paintings (T3), so E2 added these colors for analysis using the node heat map view (Fig. 1(b)). The heat map analysis results and color attribute analysis are shown in Fig. 1(a-1). From Fig. 1(a-1), it can be seen that most of these colors are related to colors of the same color system. In terms of color attributes (Fig. 1(e) and (c)), these colors are primarily

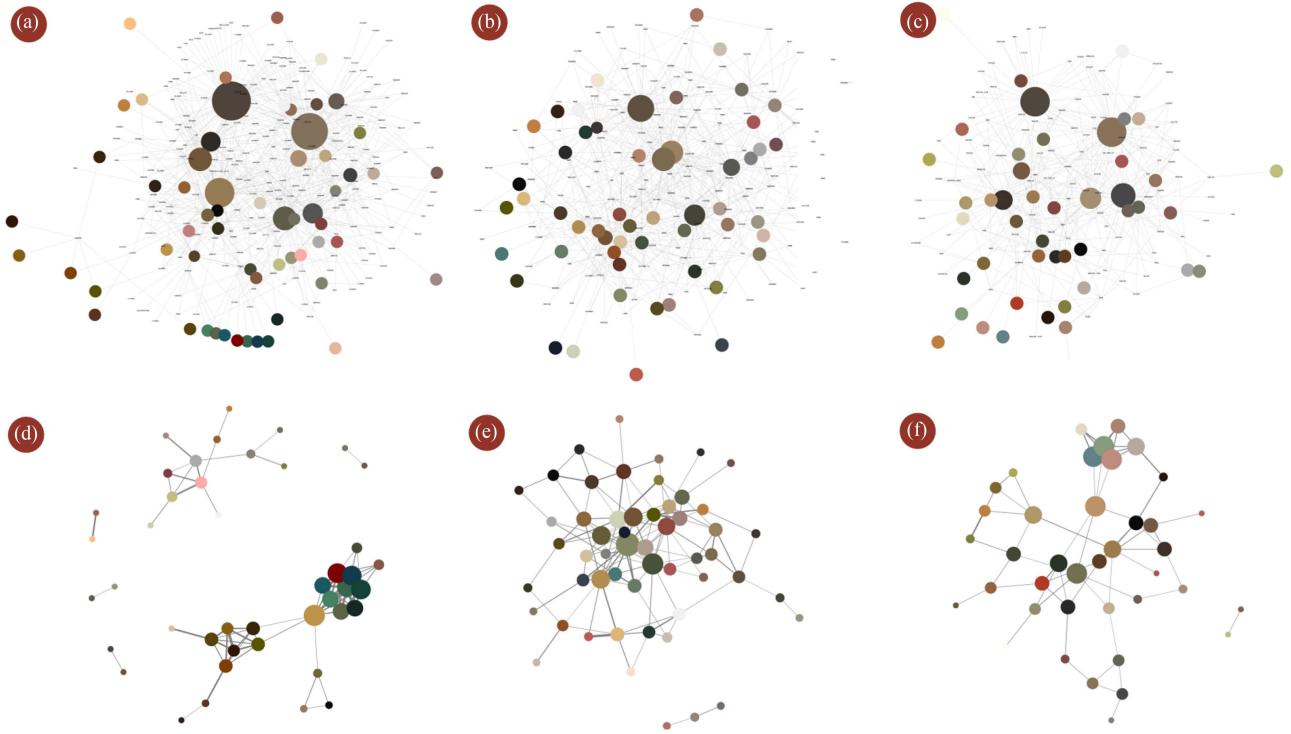


Fig. 6. Networks generated by ColorNetVis. (a)–(c) represent the color-painting bipartite networks of landscape, flower-and-bird, and figure paintings, respectively. (d)–(f) represent the One-mode color networks of landscape, flower-and-bird, and figure paintings, respectively.

from landscape paintings and flower-and-bird paintings, and the rendering methods are mostly heavy coloring and come from the Song Dynasty. Therefore, based on historical research and literature analysis, E1 and E2 summarized the colors of painting in the Song Dynasty as follows:

Cultural characteristics: Cyan and its similar colors (●, ●, ●, ●, ●, ●, ●) dominated landscape painting in the Song Dynasty, while heavy coloring were rarely used in figure painting and bird-and-flower painting, possibly related to the painting style of the time. Blue, green and cyan are colors that expresses the beauty of natural landscapes and is, therefore, widely used.

Combination colors of Cyan: In the Song Dynasty's cyan landscape paintings, gold ● was sometimes added as decoration, creating a contrast with cyan and making the painting more rich and luxurious. Red ● can form a sharp contrast with cyan, adding vitality and atmosphere to the paintings.

Case2. Historical basis for querying single color in Chinese painting colors: E3 focuses on studying individual colors for specific dynasties, hoping to investigate the performance of cyan in Song Dynasty painting. However, in reality, cyan is defined as a color between blue and green, making it difficult to understand its specific color. Therefore, E3 utilized *scope determination component*, limited the research (T1). E3 took $\Delta E = 7$ to increase the discrimination of colors and searched for the $RGB = (69, 130, 96)$ ● of cyan within the research scope, which is considered one of the standard values for 16 colors in Chinese painting. Within the scope of this study, a tree structure was generated, as shown in Fig. 1(f). In Fig. 1(f), four color nodes (●, ●, ●, ●) are close to the predefined cyan, and these

nodes generate leaf nodes with similar colors. Through the tree hierarchy diagram, E3 found that most of the cyan colors in the painting are relatively dim, and no color is completely consistent with the predefined cyan colors. Afterward, E3 generated a node attribute statistical Fig. 1(f-1) by clicking on the color nodes and found that these different turquoise colors had different statistical results (T4). After using the *node attribute statistics view*, E3 was interested in which paintings these colors appeared in. Therefore, by clicking on the painting query function in the node attribute statistics view, E3 found that there were four paintings in the database (Figs. 1(f-2) and 8). Therefore, E3 summarized the following research directions for these four paintings and these colors as follows:

Analyze the use of cyan in four paintings: Observe the specific use of cyan in four paintings, including color distribution and matching with other colors (T2, T3).

Compare cyan with other Song Dynasty paintings: Compare these four classic cases with other Song Dynasty paintings and observe whether the use of cyan in these paintings is similar. The entire visual analysis process is shown in Fig. 2(b).

Analyze the theme and artistic conception of the painting: Starting from the themes and artistic conception of these two classic cases, explore whether turquoise relates to the emotions or images the painting aims to express.

B. Researcher Reviews

To evaluate the effectiveness of ColorNetVis, we invited 7 Chinese painting color researchers (i.e., three professors (E1-E3)

and four doctoral students (E4-E7)) to participate in interviews. E1-E3 provides suggestions and guidance for the early theoretical research and multiple design iterations of ColorNetVis. Four doctoral students (E4-E7) used ColorNetVis for the first time to study painting colors during the Tang and Song dynasties. All researchers are familiar with our color database and the system used for network analysis. E4-E7 indicates that in their daily research, they mainly use painting images and color pickers to record the colors of interest and through extensive searches to obtain other information about related colors. We conducted semi-structured interviews with everyone to test whether ColorNetVis helps to explore the color composition of Chinese painting. None of these 7 participants are co-authors of this article.

System Performance: Researchers unanimously believe that the system runs smoothly, has sound visualization effects, and is organized and orderly. *This system is beneficial for formulating and analyzing color combinations in Chinese painting.* All researchers agree that their research can benefit from the proposed analytical methods. We have proposed an innovative color network analysis method, which is very helpful for researchers familiar with analyzing literature through networks to analyze colors. E3 expressed in discussing the inspiration of our system for them, extracting *unexpected* colors such as lavender. Purple was considered a color that nobles could only use in ancient China and may reveal new associations, allowing researchers like him to verify existing associations or propose new color theories. Despite favoring traditional research methods, E5 is interested in using ColorNetVis to explore some accidentally discovered colors for inspiration. E5 affirms the significance of our work and believes that ColorNetVis is a teaching system suitable for traditional Chinese color courses. In addition, E2 suggests that our method should further support multi-color comparison, which will be one of our future works.

Reliability of analysis results: Researchers must verify the color network analysis results before further studying them. E2 indicates that the color query component (Figs. 1(f) and 4(c)) is beneficial for them to understand specific colors. *The color relationships provided by ColorNetVis are reasonable enough for most of our analysis work, as the relationship between most similar colors can be discovered through color difference measurement.* As mentioned in (Section VI-A), color queries can discover nodes with similar colors in the color network and query the actual attributes of colors. E5 and E6 believe that visual design and interaction promote the connection between color and painting. E4 and E7 indicate that adding painting attributes to the display color space (Fig. 4(a)) is a good feature for checking the original material. E2-E4 points out the limitations of a single color data source and hopes to introduce more color data sources, including researchers' custom-added color data for comparing color relationships.

Universality and Scalability: In the interview, we asked which components in ColorNetVis can be directly transferred to other scenarios. Two aspects are considered valuable. 1) Color network model. E1 commented that color network visualization design is beneficial for analyzing color associations in color databases, so they should be quickly promoted to other similar

scenarios. For example, it can be applied to color analysis of Yunjin, Dunhuang murals, and traditional Chinese clothing. 2) Color nodes. E7 comment: *The nodes of the color network generated by ColorNetVis are like a classic Chinese painting color card set. I will refer to it in different research stages. It can be integrated into a new color card set for querying and inspiring designers to carry out stylized design.* In addition, E4-E7 points out that scalability may be an issue with color network visualization, as *if a network contains thousands of paintings, the network view may be very crowded.* We can avoid this problem by categorizing the attributes of the painting at more levels and providing more filtering options in the scope determination component. Multiple network hierarchies can also be used to reduce visual confusion.

C. User Study

We conducted a user study on ColorNetVis from two aspects: the workflow's effectiveness and the system's usability. We recruited 20 participants (9 females and 11 males, $age_{mean}=22.75$, $age_{sd}=2.40$) from the School of Arts and Archaeology. They have taken courses related to Chinese painting colors, are interested in the composition of Chinese painting colors, and have experience using Chinese painting elements as creative designs. We chose these participants to provide us with understandable insights and help validate the system's usability. They participate in tutorials, perform tasks, and complete questionnaires.

Each participant is required to complete two tasks. The purpose of **Task 1** is to check whether the system can help users understand the color composition of Chinese painting. In this task, participants were asked to compare the color differences closely related to cyan between landscape and flower-and-bird paintings. The purpose of **Task 2** is to allow users to explore the colors of interest in a single painting as an example. Participants conducted a free exploration system to investigate the color matching and realistic properties of colors in the *Rui He Tu* (Fig. 8) during the same historical period.

After the participants completed these two tasks, we distributed a questionnaire with questions using Likert scale criteria [43], with 1 representing "strongly disagree" and 5 representing "strongly agree". The questions in the questionnaire include four items about the effectiveness of our system workflow and four items about system's usability. We also recorded the opinions of other participants who did not participate in the questionnaire for reference, and the questionnaire results are shown in Fig. 7.

Overall, participants rated ColorNetVis highly. Regarding the effectiveness of the system workflow, participants believe that ColorNetVis can display color relationships and attributes (Q2), and they can easily explore the color composition of different types of paintings (Q3, Q4). Regarding the usability of the system, participants found that selecting an appropriate network model (Q5) based on the research content, intuitively observing and understanding the relationship between colors and paintings and color pairs (Q6), querying colors as needed (Q7), and forming a reasonable workflow (Q8) was very useful.

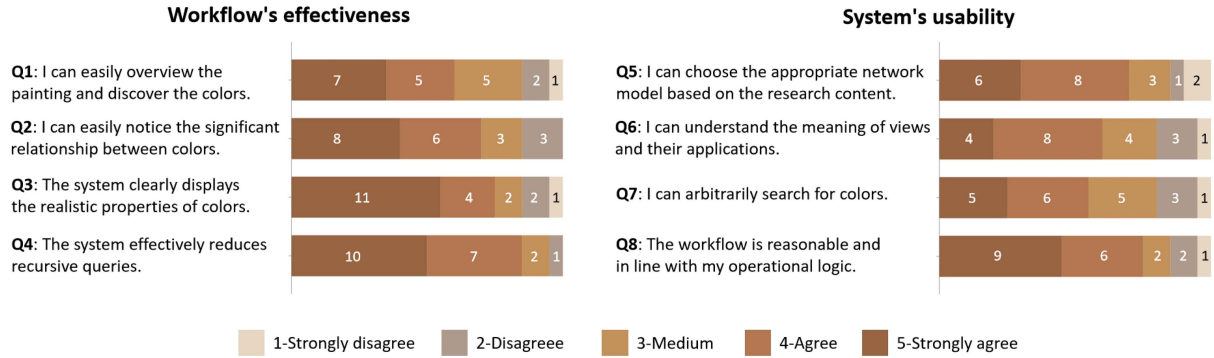


Fig. 7. Results of user perceptions of ColorNetVis. 1–5 represents “Strongly disagree” to “Strongly agree” for each statement.

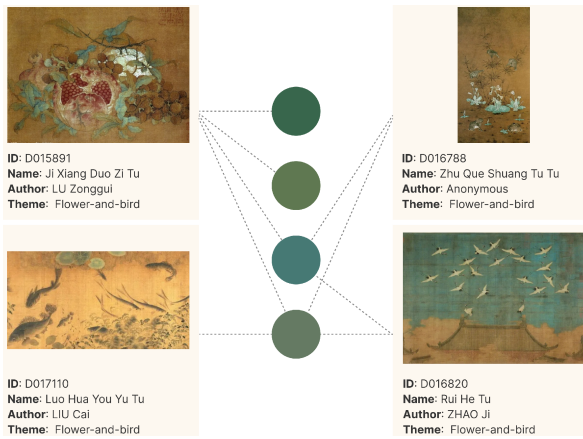


Fig. 8. Song dynasty painting corresponding to $RGB = (69, 130, 96)$.

VII. DISCUSSION

This section summarizes the benefits, limitations, and future work of ColorNetVis.

A. Benefits

System Perspective: Traditional Chinese painting is a precious “photo” that vividly depicts human history. However, for researchers of color-related digital humanities, the color composition laws and knowledge system in Chinese painting need to be systematically concluded and summarized. ColorNetVis provides a systematic color theory analysis method through a bipartite color-painting network, which can be used to describe and explain the color attens of Chinese painting in digital form. This theory can give researchers a deeper understanding of the color aesthetics and artistic value of Chinese painting.

ColorNetVis helps us study color contrast and harmony in Chinese painting through a one-mode color network. By comparing the use of colors in different paintings, we can discover the characteristics and patterns of color application in Chinese painting. We can understand how painters use these colors to express their themes through the color query and attribute analysis functions. Moreover, ColorNetVis provides a color combination analysis function, which can help researchers study the color

culture in Chinese painting history. The color application of Chinese painting has been influenced by traditional culture, with differences in materials and theme colors. By analyzing the status and role of these colors and color combinations in the network of color nodes, we can gain a deeper understanding of the cultural color styles and principles of Chinese painting. In addition, ColorNetVis establishes a digital gene for the colors of Chinese painting, explaining the complexity of color composition, thus discovering multi-attribute color patterns and obtaining new ideas and methods for color application. Therefore, ColorNetVis provides strong support for understanding and inheriting traditional art practices.

Design Meaning: Chinese painting color researchers prefer easy-to-understand and simple visualizations. Due to their habit of reading a large amount of image data of paintings, their perception of specific colors is stronger than directly presenting complex layers of colors in paintings. Our design combines paintings and colors in a bipartite network to simplify the paradigm from viewing painting images to perceiving colors. Although analyzing heat maps and explaining the centrality of nodes in color networks increase the learning costs for users, as E3 said, *We can quickly discover the relationship strength between colors on the heat map, just like giving a magnifying glass in a complex network.* Researchers value the authenticity of colors. We have identified two ways to enhance the trust and confidence of researchers in methods, processes, results, and data. The first is to provide the overall color distribution. Researchers may need clarification about the overall distribution since the color network involves merging similar colors through color differences. Therefore, we provide the original color space, which allows interactive operations to query the original painting. In addition, we provide a single color query function that can display the hierarchical structure of color difference generation, which gives researchers more confidence in the ColorNetVis process and ensures the reliability of the results.

B. Limitations and Future Work

We have outlined the limitations that need to be addressed in future work.

Data source: The current system only uses one data source. To understand diverse colors, multi-data source validation can

further enhance the flexibility of color analysis. It should be combined with other large databases or custom knowledge bases in the future. We can define more items to complete the attributes of Chinese painting.

Research on Integrated Algorithms: In network science, the performance of network visualization can be enhanced through integrated algorithms, such as link prediction [11], [19], [45], which can infer possible edges under the current network structure, i.e., discover potential color relationship pairs. Furthermore, community detection algorithm [3], [4] can identify common color-matching groups. Although these algorithms can better analyze networks, they need more interpretability. Our system has implemented interpretable basic functions and can be used to solve the problems raised. In the future, we will integrate more practical algorithms to explore color relationships from multiple perspectives.

VIII. CONCLUSION

In this work, we propose ColorNetVis, an interactive color network analysis system for researchers to identify and explore the color composition and relationships of Chinese painting. This is the first color network analysis system developed using Chinese painting color data. The color network model in ColorNetVis improves the efficiency of researchers in analyzing multiple color relationships by creating a color painting bipartite network and generating a one-mode color network through projection algorithms. ColorNetVis's visualization interface provides a color space view and individual color query components, helping researchers examine the content and associated colors of colors, cultivate trust in the system, and deepen their understanding of colors. Through case studies and interviews with researchers involved in color analysis, the effectiveness and usability of our system have been recognized.

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